

Research papers

Preferential flow reduces overland flow on slopes: insights from a field experiment on the Chinese Loess Plateau

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ABSTRACT

Preferential flow, a critical component of soil infiltration processes, plays a pivotal role in eco-hydrological dynamics. Large-scale vegetation restoration on the Chinese Loess Plateau has altered the underlying surface conditions, which in turn influences the preferential flow. However, the influence of preferential flow on slope runoff, under different vegetation restoration types and ages, remains unclear. In this study, artificial rainfall experiments, high-frequency soil moisture monitoring, and the surface covering method were used to investigate the occurrence, contribution, and influencing factors of different preferential flow patterns (non-sequential response type and high wetting-front velocity type) under varying vegetation restoration conditions. Furthermore, it elucidates the underlying mechanisms by which preferential flow influences slope runoff. The results indicate that vegetation restoration significantly increased preferential flow occurrence, which was positively correlated with vegetation age. The occurrence of preferential flow ranged from 18.9% to 40.0%, contributing 59.6% to 78.7% of the total infiltration. In revegetated land, the cumulative infiltration, initial infiltration rate, and stable infiltration rate of both matrix flow and preferential flow were significantly higher than those in cropland ($P < 0.05$). Soil structure and root characteristics were identified as the primary factors influencing preferential flow occurrence. Vegetation restoration increased preferential flow occurrence, shortened the response time between soil layers, and increased infiltration rates, reducing slope runoff. These findings provide important insights into the soil hydrological processes associated with vegetation restoration on the Loess Plateau and in other arid and semi-arid regions.

1. Introduction

Preferential flow (PF) is characterised by the nonuniform movement of water through a small fraction of the pore space, bypassing much of the soil matrix (Allaire et al., 2009; Guo & Lin, 2018). This process plays a critical role in the redistribution of soil moisture after infiltration (Gerke, 2006; Lin & Zhou, 2008). PF occurs in all soil types, especially after heavy rainfall or in low-permeability soils (Alaoui et al., 2011; Zhang et al., 2022a). Common types of PF include macropore flow, finger flow, lateral flow, and funnel flow (Allaire et al., 2009). Additional forms, such as unsaturated gravity flow, heterogeneous flow, rill

flow, oscillatory flow, and low-lying saturation are also recognised within its scope (Sidle et al., 2001). Owing to its ability to rapidly transport water and solutes, PF is a key driver of hydrological processes that affect soil erosion, slope stability, agricultural nonpoint source pollution, and various geochemical and ecological dynamics (Clothier et al., 2008). Vegetation plays a crucial role in PF by modifying soil pore structure through root growth, decay, and interactions with soil fauna, which in turn shape PF pathways (Wen et al., 2025). Vegetation restoration is a primary driver of PF and affects soil properties and root system characteristics (Qiu et al., 2022). Several studies have shown that well-structured soils can significantly increase PF occurrence (Wu et al.,

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2019). Consequently, changes in land use practices have led to different occurrences and degrees of PF.

Large-scale vegetation restoration on the Loess Plateau in China has transformed the original land use and cover patterns. Differences in vegetation types and revegetation ages have altered the humus layer, soil physicochemical properties, and root channel characteristics, which in turn affect PF processes (Gu et al., 2019; Qiu et al., 2023; Wen et al., 2025). The increased occurrence of PF on the Loess Plateau has brought new ecological challenges, including more shallow landslides, a shift in slope runoff generation from infiltration-excess overland flow to saturation-excess overland flow, and intensification of deep soil drying (Tan et al., 2017). These phenomena are linked to PF. Overland flow on slopes is not only a critical hydrological issue but also forms the foundation for soil erosion research on the Loess Plateau (Chen et al., 2018). Extensive research has investigated the factors influencing overland flow on slopes, such as rainfall intensity, slope gradient, vegetation cover, and antecedent soil moisture (Govers, 1991; Braud et al., 2001; Gu et al., 2019; Wang et al., 2022). However, research on the specific impact of PF on slope runoff remains limited (Beasley, 1976; Swarowsky et al., 2012; Kaffas et al., 2025). Moreover, the mechanisms, hydraulic properties, and morphologies of these PF pathways vary across types. Extensive research has explored the factors influencing PF, particularly soil physicochemical properties and root characteristics (Cheng et al., 2015). Soil aggregate size, distribution, and stability directly affect soil porosity, water retention, permeability, and erosion resistance, all of which shape PF behaviour (Ontl et al., 2015). Luo et al. (2010) reported that compared with other macropore parameters, factors such as the macroporosity (MP), number of macropore pathways, hydraulic radius, and the macropore angle more effectively explain water movement processes in saturated, structured soils. Plant roots are key factors influencing the development of soil PF (Xiao et al., 2024). Studies indicate that compared with soil macropores alone, root morphological characteristics contribute significantly more to PF (Perillo et al., 1999). As roots grow, they thicken, die, and decay, forming persistent tubular channels in the soil. These channels serve as essential pathways for PF (Cai et al., 2024). Furthermore, interconnected root networks create a continuous three-dimensional macropore system, significantly enhancing soil pore connectivity and enabling water to move through a more tortuous yet efficient pore network (Zheng et al., 2023). In addition, root exudates and organic substances derived from rhizosphere microbial activity promote soil aggregate formation, which helps stabilise the structure around root channels, prevent pore collapse, and thereby maintain the long-term stability of PF paths (Zhang et al., 2023a). These findings are supported by those of Benegas et al. (2014), who confirmed that plant roots play a vital role in increasing soil macroporosity, connectivity, aggregate content, and soil organic matter, thereby facilitating the development of PF. Nevertheless, comprehensive studies examining the combined factors influencing PF remain scarce.

In recent years, quantitative studies of PF have expanded across multiple scales and methods. Traditional approaches for studying PF, such as dye tracing and breakthrough curve methods (Allaire et al., 2009; Wuest, 2009; Beven & Germann, 2013; Zhang et al., 2015), are valuable but often invasive, and limited in their ability to capture real-time dynamics (Wielenkamp et al., 2016). In response, nondestructive and long-term monitoring techniques such as tomography, ground-penetrating radar, and thermal imaging have been introduced (Sammartino et al., 2012; Zhang et al., 2014). While promising, these methods are constrained by their low resolution, difficulties in directly measuring certain parameters, and uncertainties in inversion algorithms (Lin & Zhou, 2008; Angermann et al., 2017). High-frequency, high-resolution in situ soil moisture monitoring has since emerged as a more effective approach for capturing PF dynamics. This method is widely used because of its low cost, operational simplicity, and minimal disturbance to the natural soil environment during installation (Guo & Lin, 2018). Moreover, it is suitable for both natural and simulated rainfall conditions (Hardie et al., 2013), offering high-density and high-

frequency soil moisture measurements across varying soil depths to detect PF processes at the microscopic level. On the basis of soil moisture sensor monitoring data, four soil moisture response patterns associated with infiltration events were identified: (1) non-sequential response time type, (2) high wetting front velocity type, (3) sequential response type, and (4) non-response type. Among these, the non-sequential response time type and the high wetting front velocity type are considered indicators of PF (Germann & Hensel, 2006; Hardie et al., 2013; Zhang et al., 2022; Nimmo et al., 2025). Numerous studies have used soil moisture time series data to identify PF events and investigate their relationships with spatial variability and hydrological conditions (Koestel & Jorda, 2014; Kaffas et al., 2025; Nimmo et al., 2025). However, most related research has focused on natural rainfall events, with few studies utilising artificial rainfall simulations. Furthermore, research has shown that vegetation cover and slope location significantly influence the types and occurrence of PF (Tang et al., 2020; Wu et al., 2022). Nonetheless, studies integrating small-scale soil properties, root characteristics, and PF types across different vegetation covers, particularly those using high-frequency, high-resolution soil moisture monitoring under natural initial and boundary conditions remain scarce. Moreover, although the importance of PF has been widely recognised, most studies have primarily focused on its type, occurrence, and responsiveness to rainfall events (Graham & Lin, 2011; Wielenkamp et al., 2016; Zhang et al., 2022), while limited attention has been given to soil moisture recession patterns under different infiltration patterns and vegetation restoration types. This gap is critical, as analysis of the post-rainfall soil moisture recession curve is essential for the calibration and validation of hydrological models (Romero et al., 2017; Basso et al., 2021).

While numerous studies have explored PF mechanisms and their impacts, understanding the specific contributions of PF relative to matrix flow (MF) to infiltration has been a critical focus of recent experimental research (Sanders et al., 2012; Zhang et al., 2017; Zhang et al., 2019a, Zhang et al., 2022b). Zhang et al. (2019a) quantified the contribution of PF using two double-ring infiltrometers, one to measure total infiltration and the other to measure MF by blocking macropores with sandy soil. Zhang et al. (2022b) introduced a new technique, the double infiltration method, which combines field-based double-ring infiltrometers with laboratory-based constant-head permeameters to further quantify PF. Sanders et al. (2012) used liquid latex to seal macropores at the soil surface and assessed their contribution to total flow under various field conditions. Building on these methods, our experiment quantified the PF contribution to both infiltration and slope runoff by sealing macropores and PF pathways with a sand–soil mixture, which we refer to as the slope preferential flow method. Several studies have also documented the consistency of soil matrix infiltration and hydraulic properties, such as saturated hydraulic conductivity, in neighbouring areas (Sanders et al., 2012; Müller et al., 2018). We also considered the effects of the sand–soil mixture and gauze (water flow is allowed to pass, but the sand–soil mixture is not allowed to pass) on the experimental outcomes during the rainfall simulation.

Given the critical role of PF in soil infiltration and slope runoff processes, as well as the limited quantitative research on PF, this study, set within the context of vegetation restoration on the Chinese Loess Plateau, aimed to investigate the mechanisms of PF generation and its impact on slope runoff. Through field-based artificial rainfall simulations paired with high-frequency, high-resolution soil moisture sensors, this study addressed three primary objectives: 1) to quantitatively distinguish the contribution of PF to total infiltration; 2) to identify the occurrence of preferential flow on the basis of soil moisture response patterns, specifically non-sequential response and high wetting-front velocity under different vegetation restoration types and restoration ages; 3) to clarify the factors influencing PF occurrence; and 4) to examine the effects of PF on soil moisture depletion and slope runoff.

2. Materials and methods

2.1. Study area

The experiment was conducted in the Zhifanggou catchment ($36^{\circ}42' - 36^{\circ}46'N$, $109^{\circ}13' - 109^{\circ}16'E$) in the central Yanhe River catchment, located on the Loess Plateau, China (Fig. 1). This catchment covers an area of 8.275 km^2 and has a narrow north-south orientation. This region is representative of the typical hilly and gully landscape of the Loess Plateau. The study area receives an average annual precipitation of 505 mm, with significant interannual variability ranging from 300 mm in dry years to over 700 mm in wet years. Precipitation is unevenly distributed throughout the year, with 74% of the total occurring between June and September, primarily in the form of intense storms. The region has a frost-free period of 158 days and an average annual temperature of 8.8°C (Qiu et al., 2022). The soils in the study area are homogeneous, forming a transition zone between loessial soil and sandy loess. The predominant zonal soil is black loam, which is classified as entisols according to the USDA Soil Taxonomy. These soils are characterised by a loose texture, low erosion resistance, and high susceptibility to water and soil loss. Due to effective soil and water conservation strategies and land use adjustments, the vegetation in the catchment has significantly improved. The landscape now consists of forests, shrubs, and grasses. The forestlands are primarily dominated by *Robinia pseudoacacia* and *Populus simonii*, whereas the shrublands are dominated by *Caragana korshinskii* and *Hippophae rhamnoides*. Grasslands are mainly composed of *Artemisia gmelinii*, *Artemisia giraldii*, *Stipa bungeana*, and *Bothriochloa ischaemum* (Wang et al., 2014).

2.2. Experimental design and treatments

2.2.1. Selection of experimental plots and artificial rainfall devices

The space-for-time substitution method was used to select the experimental plots within the study area. The core principle of this method is to infer the dynamic changes in the time dimension by analysing the differences in the spatial dimension, that is, to reflect the evolution of the time series by using the differences in the spatial position (Walker et al., 2010). Six plots representing the major vegetation restoration types were selected within a 5-km radius (Table 1). These plots included forestlands with *Robinia pseudoacacia* established for 20 (F20) and 40 years (F40); shrublands dominated by *Caragana korshinskii* growing for 20 (S20) and 40 years (S40); and grasslands with dominant species such as *Artemisia gmelinii*, *Artemisia giraldii*, *Stipa bungeana*, and *Bothriochloa ischaemum* growing for 20 (G20) and 40 years (G40). A cropland plot under maize cultivation was selected as the control (Table 1). To reduce potential topographic effects, experimental plots were chosen within narrow ranges of slopes, aspects, and elevations during site selection. As a result, the topographic variability among the plots was minimal. All the plots were formerly cropland before they were converted to their current land use types, ensuring comparability of their land use types (Gu et al., 2019).

A custom-made portable needle-type rainfall simulator was used for the field rainfall simulation experiments. This device is widely utilised because of its portability, ease of data collection, and high degree of rainfall uniformity (Navas et al., 1990; Zhao et al., 2013; Grismer, 2014; Zhang et al., 2019b). The system comprises a rainfall generation unit, a constant water supply system, and a mobile support structure. The

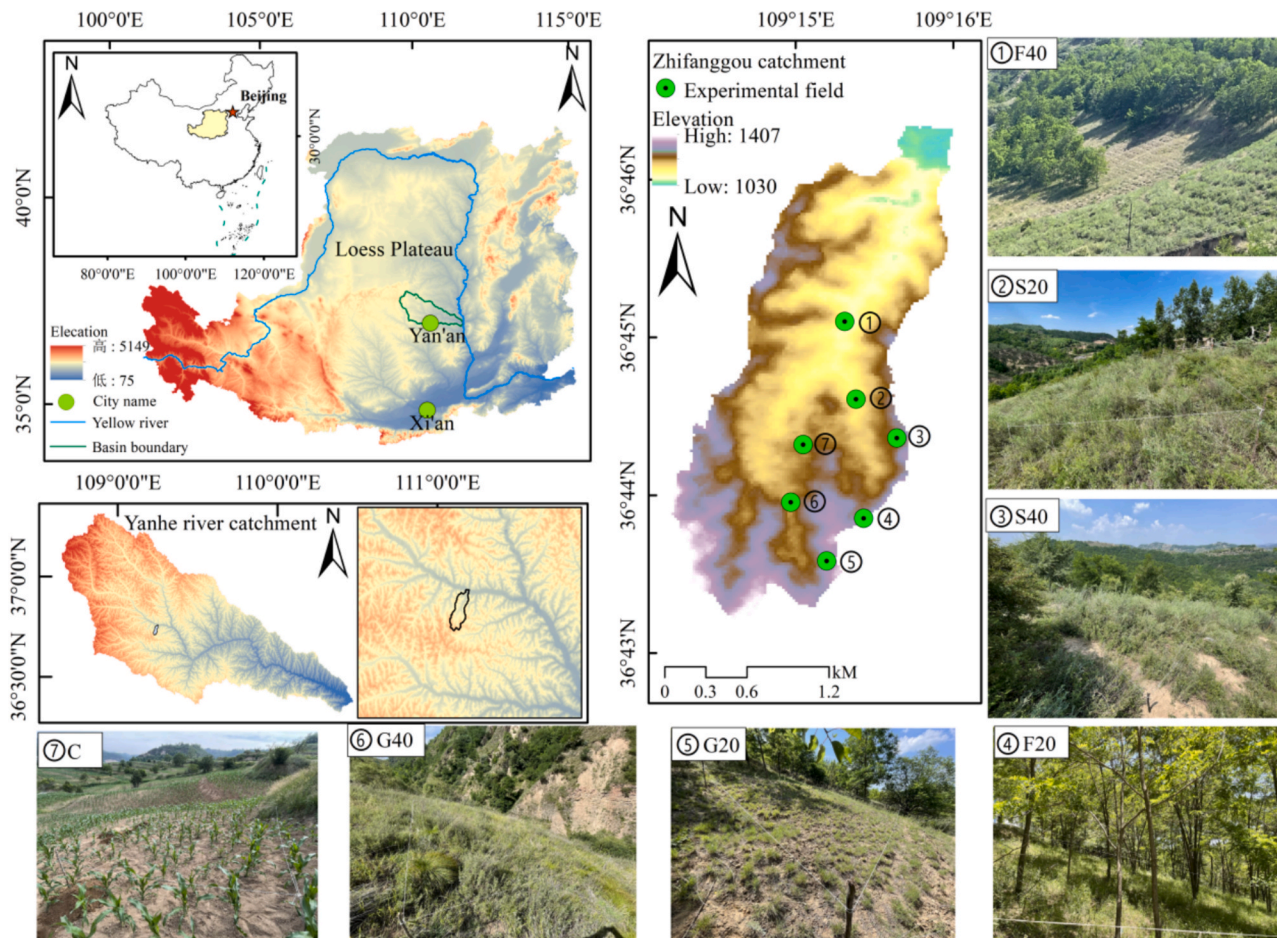


Fig. 1. The information and photos of the study area and experimental treatments. C represents cropland, G20 represents 20-year grassland, G40 represents 40-year grassland, S20 represents 20-year shrubland, S40 represents 40-year shrubland, F20 represents 20-year forestland, F40 represents 40-year forestland.

Table 1

Basic information of sampling plots.

Land use	Vegetation	Elevation (m)	Aspect	Slope (°)	Slope location	Vegetation coverage (%)	Year	Intensity (mm min ⁻¹)
Forestland	RP	1273	north-facing	16.67(1.12)	middle	50(1)	20	2.01(0.12)
Forestland	RP	1106	north-facing	17.54(2.13)	middle	61(2)	40	2.03(0.14)
Shrubland	CK, SD	1171	north-facing	16.33(2.89)	middle	53(2)	20	2.05(0.04)
Shrubland	CK, SD	1263	north-facing	16.45(1.47)	middle	60(2)	40	2.02(0.21)
Grassland	AG, AC, BI, AA, SB	1337	north-facing	17.85(2.24)	middle	44(2)	20	1.99(0.17)
Grassland	AG, AC, BI, AA, SB	1220	north-facing	17.33(1.21)	middle	61(2)	40	2.04(0.08)
Cropland	Corn	1214	north-facing	15.33(0.78)	middle	17(1)	—	2.02(0.06)

RP: Robinia pseudoacacia; CK: Caragana korshinskii; SD: Sophora davidii; AG: Artemisia gmelinii; AC: Artemisia capillaris; BI: Bothriochloa ischaemum; AA: Astragalus adsurgens; SB: Stipa bungeana. Outside the brackets is the mean, and within the brackets is the standard deviation.

rainfall generation unit, measuring $1.6 \text{ m} \times 0.8 \text{ m} \times 0.3 \text{ m}$, features 512 (32×16) syringe needles at its base. Rainfall intensity is regulated by adjusting the height of the constant water head and the needle size, simulating rainfall intensities ranging from $0.5 \text{ mm} \cdot \text{min}^{-1}$ to $3.0 \text{ mm} \cdot \text{min}^{-1}$ (Gu et al., 2019). The constant water supply system maintains the water head height via a valve, and the mobile support structure enables a raindrop fall height exceeding 2 m. To ensure a uniform rainfall panel across varying slopes, the drop height was standardised to 1.5 m. Given the constraints of the rainfall simulator, sensor installation, and field conditions, we chose a runoff plot size of $1.5 \text{ m} \times 0.6 \text{ m}$ for the experiments. Each runoff plot was enclosed by rigid steel plates that were 25 cm wide and embedded 15 cm into the ground to establish plot boundaries (Fig. 2a). The plot slope angles ranged from 15.33° to 17.85° , with minimal variation, rendering slope effects negligible for the experiment. For each runoff plot, three vertical vegetation photographs were captured using a digital camera (Nikon D7500, Japan), and vegetation coverage was quantified using ENVI software (Zhu et al.,

2023).

2.2.2. Design of different artificial rainfall experiments

To ensure that runoff was generated in all the plots, the rainfall intensity for the experiments was set at $2 \text{ mm} \cdot \text{min}^{-1}$. The experiments were conducted in two stages to monitor and quantify the preferential flow (PF). Stage one entailed monitoring the occurrence of PF and the characteristics of slope runoff under different vegetation restoration types and revegetation ages; Stage two subsequently quantified the PF contribution by isolating matrix and preferential flow.

In the first stage, the surface vegetation in each runoff plot was retained, and a total of 21 rainfall experiments were conducted (7 plots $\times$ 3 replicates). Frequency-domain reflectometry (FDR) probes (JXBS-3001-TR, China) with a resolution of 0.1% were installed at upslope (25 cm), mesoslope (75 cm), and downslope (125 cm) locations, on both sides of the runoff plot. Soil moisture sensors were installed at depths of 10, 20, 30, 40, and 50 cm from the surface in the vertical direction at

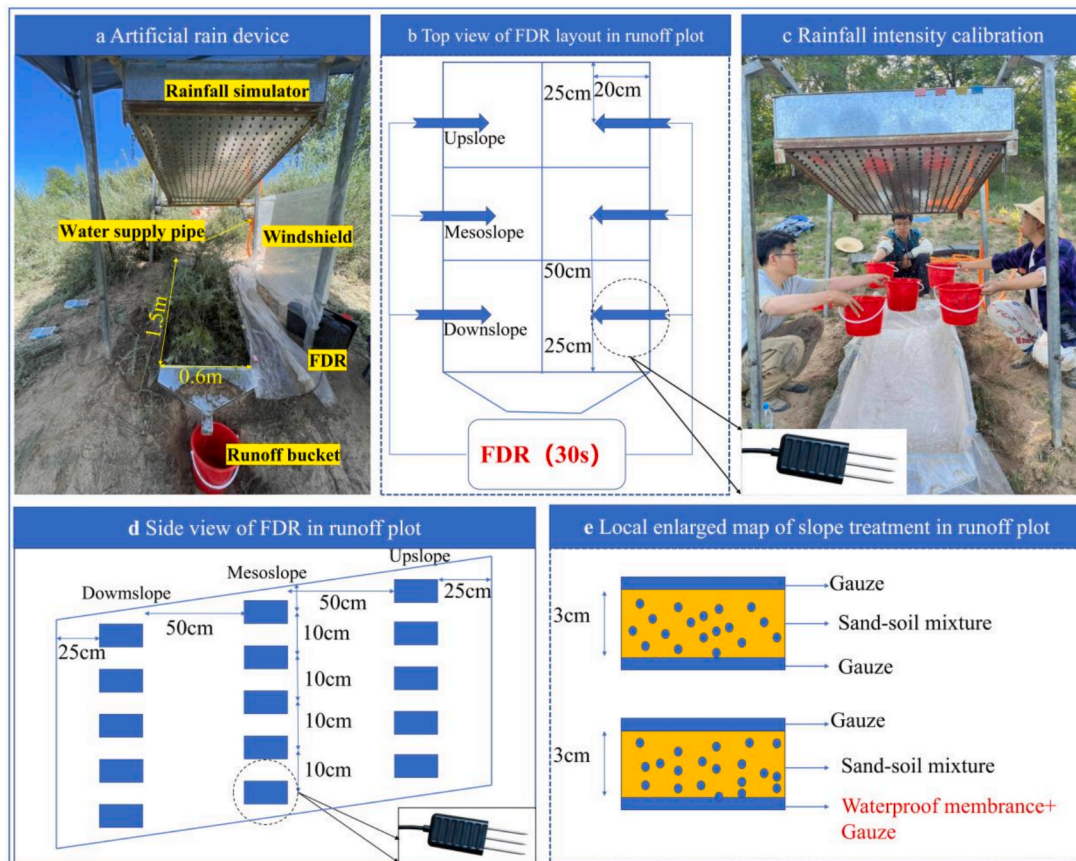


Fig. 2. Field rainfall experiment setup and treatment. a is the information and photos of the rainfall experiment device, b and d are the detailed arrangement of FDR probes in the soil, c is the rainfall intensity calibration process and photos, e is the slope treatment process for quantitatively distinguishing PF and MF.

each slope location, for a total of 30 soil moisture sensors (2 sides $\times$ 3 slope locations $\times$ 5 verticals). These sensors recorded data at 30-second intervals to track dynamic changes in soil moisture during rainfall. The sensors were installed 20 cm from the plot edges and monitored a soil volume of 0.015 m³ (0.3 m $\times$ 0.5 m $\times$ 0.1 m), which increased the likelihood of detecting PF (Fig. 2b, d). After installation, the original soil was carefully backfilled into the pits and compacted to restore its original bulk density and minimise disturbance.

The second stage focused on four experimental plots: F40, S40, G40, and cropland. Three runoff plots with the same horizontal orientation were established in each plot, and surface vegetation, litter, and crust were removed. The runoff plots were treated as follows, from left to right:

- (1) Direct rainfall was applied to measure the total infiltration of both PF and matrix flow (MF) (Fig. S1a).
- (2) Permeable but sandproof gauze was laid on the slope surface, followed by a 3 cm layer of sand–soil mixture adjusted to match the bulk density (BD). A second layer of gauze was added on top to prevent rainwater from washing away the sand–soil mixture. This treatment was used to measure MF infiltration and runoff generated by the gauze and sand–soil mixture (Fig. S1b).
- (3) A waterproof membrane was first laid on the slope surface, followed by a gauze, a 3-cm layer of sand–soil mixture, and a final gauze layer on top. This treatment was designed to measure the runoff generated solely by the gauze and sand–soil mixture (Zhang et al., 2019a) (Fig. S1c). Perroux and White (1988) and Kodesová et al. (2012) suggested that sand–soil mixtures, as cover materials, have slightly higher saturated hydraulic conductivity (K_s) than the soil matrix does and can effectively block PF. A higher soil proportion enhances the shielding effect on PF. To measure the K_s of the soil matrix, disturbed soil samples were collected, processed through grinding and sieving, and tested for the K_s of surface soil (0–20 cm) from all the experimental plots. The K_s of cropland, 20-year grassland (G20), 40-year grassland (G40), 20-year shrubland (S20), 40-year shrubland (S40), 20-year forestland (F20), and 40-year forestland (F40) were 0.43 mm·min^{−1}, 0.64 mm·min^{−1}, 0.69 mm·min^{−1}, 0.67 mm·min^{−1}, 0.77 mm·min^{−1}, 0.68 mm·min^{−1}, and 0.73 mm·min^{−1}, respectively. Moreover, the K_s of sand–soil mixtures with different ratios of 1:1, 1:2, 2:3, 1:3, 1:4, and 1:5 were measured, with values of 2.15 mm·min^{−1}, 1.37 mm·min^{−1}, 0.84 mm·min^{−1}, 0.75 mm·min^{−1}, 0.44 mm·min^{−1}, and 0.27 mm·min^{−1}, respectively. A 1:4 sand–soil mixture was chosen for the cropland; a 1:3 sand–soil mixture for G20, G40, S20, F20, and F40; and a 2:3 sand–soil mixture for S40 as the cover material. In this stage, a total of 36 rainfall events were conducted (4 plots $\times$ 3 treatments $\times$ 3 replicates).

The principle of this method involves simultaneously establishing three runoff plots of identical size within a site under the same vegetation type, with each plot spaced 1 m apart. In the first runoff plot, the overland area was covered with a sand–soil mixture and gauze to measure the matrix flow infiltration process. The second runoff plot remained untreated to measure the total infiltration process. In the third runoff plot, only gauze was applied to the overland area to determine the increase or decrease in slope runoff caused by the presence of gauze. On the basis of these measurements, the preferential flow infiltration process can be quantitatively isolated. Although no previous literature directly supports this specific method for calculating PF, its underlying physical mechanisms are well known. We assumed that, under the same vegetation type, the soil properties and structure were similar among the three runoff plots and that the infiltration rate of the soil matrix flow was identical (Zhang et al., 2017, 2019a). The formulas used for these measurements are as follows:

$$I_m + I_p = I_{r1} - R_a \quad (1)$$

$$I_m = I_{r2} - R_b - R_s \quad (2)$$

$$R_s = I_{r3} - R_c \quad (3)$$

where I_m represents the matrix infiltration rate (mm·min^{−1}), I_p represents the PF infiltration rate (mm·min^{−1}), I_{ri} represents the rainfall intensity (mm·min^{−1}) ($i = 1, 2, 3$), R_i represents the runoff rate under different treatments (mm·min^{−1}) ($i = a, b, c$), and R_s represents the reduction in the runoff rate caused by the gauze and sand–soil mixture (mm·min^{−1}).

2.2.3. Measurements and data analysis

Rainfall experiments were conducted from June to August 2023 and from June to July 2024. This timing was chosen to both align with the natural rainy season—ensuring that the artificial rainfall conditions were representative—and to capture the period of peak vegetation activity. During these months, vegetation (particularly corn) is at its most vigorous stage of growth, consequently exerting the greatest influence on soil properties and root characteristics, and promoting the most active period of preferential flow. To minimise errors due to the initial soil moisture content, a prewetting rainfall with an intensity of 2 mm·min^{−1} was applied 24 h before the first rainfall event in all the runoff plots to initiate runoff. After each experiment, the plots were exposed to natural conditions for 24 h before the next rainfall event. If natural rainfall occurred during this period, the exposure time was extended accordingly. Before each experiment, the plots were covered with waterproof sheets. Rainfall was collected from five positions within each plot to ensure that the measured rainfall intensity across these positions did not deviate by more than 10% from the target intensity (Gu et al., 2019) (Fig. 2c). Once the rainfall intensity was calibrated, the waterproof sheets were promptly removed, and the timing began. Runoff was recorded as soon as it reached the collection trough. Each rainfall event lasted for 1 h. Runoff samples were collected using runoff buckets at 1-minute intervals from 0 to 20 min after runoff began, and then every 2 min until runoff ceased.

During Stage One, the initial runoff time, post-rainfall runoff time, cumulative runoff, and post-rainfall cumulative runoff were recorded. Soil moisture sensors continuously monitored soil moisture changes for 24 h after the rainfall experiment, after which all sensor data were exported. For each sensor, the response time, difference in response time between adjacent soil layers, initial soil moisture content before rainfall, maximum soil moisture content, stable soil moisture content, and soil moisture recession were recorded (Zhang et al., 2022a) (Fig. S2). During Stage Two, the total runoff, infiltration volume, initial infiltration rate, and stable infiltration rate were measured under the different treatments. On the basis of continuous measurements, the average infiltration rates during the first three and last ten minutes were used to represent the initial infiltration rate and stable infiltration rate, respectively (Qiu et al., 2023). A response event is considered to begin when the soil moisture increases beyond the instrument noise level (Hardie et al., 2013). For sensors with an accuracy of 1% to 4%, a soil moisture change must exceed 0.2% to be considered significant (Blume et al., 2009; Germann & Hensel, 2006). In this experiment, the minimum threshold for soil moisture change was set at 0.3%, indicating that a sensor response was recorded when the soil moisture increased by 0.3%.

2.2.4. Identification of infiltration patterns

On the basis of the response patterns of the soil moisture sensors along the same vertical direction, the infiltration events were: non-sequential response (NSR), high wetting-front velocity (HWFV), sequential response (SR), and non-response (NR). Multiple infiltration response patterns associated with infiltration events may occur within a single rainfall event in each runoff plot. Among these, the NSR and

HWFV are both indicative of preferential flow. The response time sequences of the soil moisture sensors were categorised into the following soil moisture response patterns used to identify PF:

- (1) NSR is defined as a soil moisture response pattern in which, along the same vertical profile, at least one FDR sensor responds earlier than a sensor positioned at a shallower depth does (Hardie et al., 2013).
- (2) HWFV is defined as a soil moisture response pattern in which, along the same vertical profile, the response time of the FDR sensors increases progressively from top to bottom, and the maximum velocity of the wetting front (V_{max}) between adjacent sensors is greater than the K_s of the soil layer; $V_{max} = \Delta h / DRT$, where Δh represents the distance between adjacent soil layers (10 cm in this study), and DRT represents the difference in response time between adjacent soil layers. (Germann & Hensel, 2006).
- (3) SR, used as a soil moisture response pattern, is characterised by progressively increasing response times of frequency domain reflectometry sensors with depth along the same vertical profile, with the V_{max} between adjacent sensors being lower than the K_s of the soil layer.
- (4) NR, used as a soil moisture response pattern, is defined by a change in soil moisture of less than 0.3% detected by the FDR sensor throughout the monitoring period.

The occurrence of each infiltration pattern was calculated using the following formula:

$$NSR/HWFV/SR(\%) = \frac{\text{the number of NSR/HWFV/SR responses}}{\text{the total number of FDR}} \quad (3)$$

2.3. Soil moisture recession curve

To compare and analyse the differences in the effects of different preferential flow types on the soil moisture recession process, the soil moisture recession curve is introduced (Rutledge & Mesko, 1996). This curve represents the rate at which soil moisture is lost from saturated soil under natural conditions, as shown in the following formula:

$$\theta_t = \theta_0 K^t \quad (5)$$

where θ_0 represents the soil water content at a certain time, t represents the time interval (h), θ_t represents the soil water content after t h, and K is the recession constant.

K reflects the soil's water retention capacity. A larger K value indicates slower moisture depletion and a greater soil water retention capacity (Rivera-Ramírez et al., 2002). The most commonly used method for estimating K is through the master recession curve (MRC) approach. The specific calculation process is described in the study by Sujono et al. (2004). The accuracy of the K value is related to the time interval t . The larger t is, the higher the accuracy of K , but if t is too large, it may exclude certain recession segments from the data cycle. After comprehensive consideration, this study employed a 3-hour time interval (Zhang et al., 2022a).

2.4. Collection and measurement of soil and roots

To investigate the factors influencing runoff and preferential flow, we measured soil physicochemical properties and root characteristics. In each runoff plot, 105 disturbed soil samples (seven vegetation restoration types $\times$ five soil layers $\times$ three replicates) were collected using 100 cm³ ring cutters and 2000 cm³ stainless steel boxes at 10 cm intervals. These samples were analysed to determine the bulk density (BD), particle size distribution, the content of soil aggregate larger than 0.25 mm (AGC), and soil organic matter content (SOM). Undisturbed soil samples were also collected with 100 cm³ ring cutters for subsequent macropore

analysis. For root analysis, 105 root samples (seven vegetation restoration types $\times$ five soil layers $\times$ three replicates) were obtained by mixing soil samples from the same layers. These samples were collected with a 9-cm diameter root auger from undisturbed areas near the runoff plots at 10-cm intervals.

The BD and initial moisture content were determined by the oven-drying method. SOM was assessed using the potassium dichromate oxidation method, and the pH was measured with a pH meter at a soil-to-water ratio of 2.5:1. The particle size distribution was analysed with a Mastersizer 2000 laser particle size analyser (Malvern Instruments Ltd., Malvern, UK), and the soil particles were classified into sand (2.0–0.05 mm), silt (0.05–0.002 mm), and clay (< 0.002 mm) fractions according to the USDA soil classification system. The AGC was determined by the wet sieving method. Macropores were defined as voids with diameters greater than 0.047 mm and were identified with a nanoVoxel1280 industrial computed tomography scanner set to a peak voltage of 60 kV and a current of 0.6 mA. The scan generated images with a resolution of 0.047 mm and a pixel matrix of 1280 $\times$ 1280 pixels, covering an area of 128 mm $\times$ 128 mm. MP, connectivity (Con), and fractal dimension (FD) were calculated using Avizo3D (2020) software from slice images (Houston et al., 2017). The root length density (RLD, cm $\cdot$ cm⁻³), root surface area density (RSD, cm² $\cdot$ cm⁻³), and root volume density (RVD, cm³ $\cdot$ cm⁻³) were measured with an Epson V900 photo scanner (Japan). The roots were then oven-dried at 60 °C for 72 h to determine the root weight density (RWD, mg $\cdot$ cm⁻³) with an accuracy of 0.001 g. The following formulas were used:

$$RLD = RL/V \quad (6)$$

$$RSD = RS/V \quad (7)$$

$$RVD = RV/V \quad (8)$$

$$RWD = RW/V \quad (9)$$

where RL is the total root length (cm), RS is the total root surface area (cm²), RV is the total root volume (cm³), RW is the root dry weight (mg), and V is the soil volume from which roots were extracted for each layer (2000 cm³).

2.5. Statistical analysis

The effects of vegetation restoration types on runoff characteristics, including initial runoff time, cumulative runoff, post-rainfall runoff time, and post-rainfall cumulative runoff, were assessed using one-way analysis of variance. Cumulative infiltration of preferential flow, the proportion of cumulative infiltration of preferential flow, the initial infiltration rate of preferential flow, and the stable infiltration rate of preferential flow were also evaluated using one-way analysis of variance. Additionally, the influence of slope location, vegetation restoration, and infiltration pattern on soil moisture depletion was analysed by factorial analysis of variance. Normality was tested using the Kolmogorov–Smirnov test, and non-normal data were log-transformed (base 10) to meet normality assumptions. Variance homogeneity was tested with Levene's test. Pearson's correlation and linear regression were used to investigate the relationships between PF occurrence and runoff as well as infiltration. Additionally, partial least squares path modelling (PLS-PM) was performed to explore the causal relationships between the non-sequential response, high wetting-front velocity, and their influencing factors, including the content of aggregates larger than 0.25 mm, soil macroporosity, macropore connectivity, root length density, root surface area density, root volume density, and root weight density. The suitability of the model was assessed using a goodness-of-fit test. All statistical analyses were performed using IBM SPSS Statistics 26, PLS-PM was conducted using the "plspm" package (Sanchez et al., 2017) in R version 4.0.3 (R Core Team 2020).

3. Results

3.1. Characteristics of slope runoff

All slope runoff patterns across different vegetation restoration types initially sharply increased, followed by stabilisation and an eventual rapid decrease to zero (Fig. 3). Compared with cropland, revegetated land had significantly lower runoff rates, with greater revegetation age correlated with lower runoff rates. Revegetated lands also exhibited delayed initial runoff times, extended post-rainfall runoff times, but reduced cumulative runoff and post-rainfall cumulative runoff ($P < 0.01$) (Table 2). Specifically, compared with cropland, cumulative runoff decreased by 58.70%, 65.17%, 62.93%, 74.27%, 75.60%, and 68.47% in 20-year grassland, 40-year grassland, 20-year shrubland, 40-year shrubland, 20-year forestland, and 40-year forestland, respectively, and the initial runoff time increased by 171.34 s, 222.00 s, 228.34 s, 344.67 s, and 380.00 s, respectively. Among these areas, shrubland exhibited the highest water retention.

3.2. Occurrence of PF and its contribution to infiltration

3.2.1. Occurrence of different infiltration patterns

After the conversion of cropland to revegetated land, the occurrence of preferential flow (PF) increased significantly (Fig. 4). PF occurrence, identified on the basis of the combined frequencies of non-sequential response (NSR) and high wetting-front velocity (HWFV), was significantly greater in revegetated land than in cropland ($P < 0.01$) and increased with vegetation restoration age (Fig. 4a). PF occurrences in cropland, 20-year grassland, 40-year grassland, 20-year shrubland, 40-year shrubland, 20-year forestland, and 40-year forestland were 18.9%, 25.6%, 33.3%, 21.1%, 37.8%, 24.4%, and 40.0%, respectively. HWFV occurred 1.72 to 2.40 times more frequently than NSR did. PF occurrence was highest on the upslope (32.9%), followed by the meso-slope (27.5%) and downslope (27.1%), but the differences were not significant ($P = 0.18$) (Fig. 4b). However, PF occurrence decreased significantly with increasing soil depth ($P < 0.01$) (Fig. 4c). Among the interactions, only that between slope location and soil depth had a significant effect on PF occurrence ($P < 0.05$) (Table S1). In summary, vegetation restoration significantly increased PF occurrence. While the variation in PF occurrence across slope locations was not significant, higher frequencies tended to occur upslope compared than that other slope locations.

3.2.2. Contribution of PF to infiltration

The contribution of preferential flow (PF) to total infiltration ranged

from 59.59 to 78.70% (Table 3). Compared with cropland, revegetated land resulted in significantly greater cumulative infiltration of matrix flow (MF), initial infiltration rate of MF, stable infiltration rate of MF, cumulative infiltration of PF, initial infiltration rate of PF, and stable infiltration rate of PF ($P < 0.05$). Among the different land use types, S40 had the highest cumulative infiltration of PF (57.37 mm), while F40 had the highest proportion of cumulative infiltration of PF (78.70%). F40 also resulted in the highest initial infiltration rate of PF (1.33 mm·min⁻¹) and the largest proportion of initial infiltration rate of PF (66.50%). With respect to the stable infiltration rate of PF, S40 had the highest rate (0.82 mm·min⁻¹), while F40 had the greatest proportion of stable infiltration rate of PF (78.55%). In summary, vegetation restoration significantly enhanced the soil PF infiltration capacity, with longer revegetation ages leading to greater PF infiltration. Accordingly, F40, which represented the oldest revegetation age, exhibited the highest PF infiltration capacity.

3.3. Factors influencing PF occurrence

Root characteristics (root length density (RLD), root surface area density (RSD), root volume density (RVD) and root weight density (RWD)) and soil properties (soil macroporosity (MP), macropore connectivity (Con), and the content of aggregates larger than 0.25 mm)) were the main factors affecting the occurrence of preferential flow (Fig. 5). PF occurrence identified by non-sequential response (NSR) significantly positively correlated with the AGC and MP ($P < 0.05$), and with the Con, RLD, RSD, RVD, and RWD ($P < 0.01$). PF occurrence identified by high wetting-front velocity (HWFV) significantly positively correlated with the Con and RWD ($P < 0.05$), and with the AGC, RLD, RSD, and RVD ($P < 0.01$). Furthermore, the partial least squares path modelling results showed that the model explained 75% and 78% of the variance in the NSR and HWFV, respectively (Fig. 6). Plant root characteristics (i.e., RLD, RSD, RVD, and RWD) had the strongest and most direct effect on the NSR ($\beta = 0.49$), followed by the MP ($\beta = 0.25$), with the smallest effect observed for the AGC ($\beta = 0.11$). Plant root characteristics remained the strongest and most direct factor influencing the HWFV ($\beta = 0.45$), followed by the AGC ($\beta = 0.43$), with the Con having the smallest effect ($\beta = 0.16$). Thus, plant root characteristics had the most significant and direct effect on PF occurrence. In addition to root characteristics, the MP was a key factor influencing the NSR, whereas the AGC was the dominant factor influencing the HWFV.

3.4. Soil moisture recession

The effects of infiltration events identified by different soil moisture

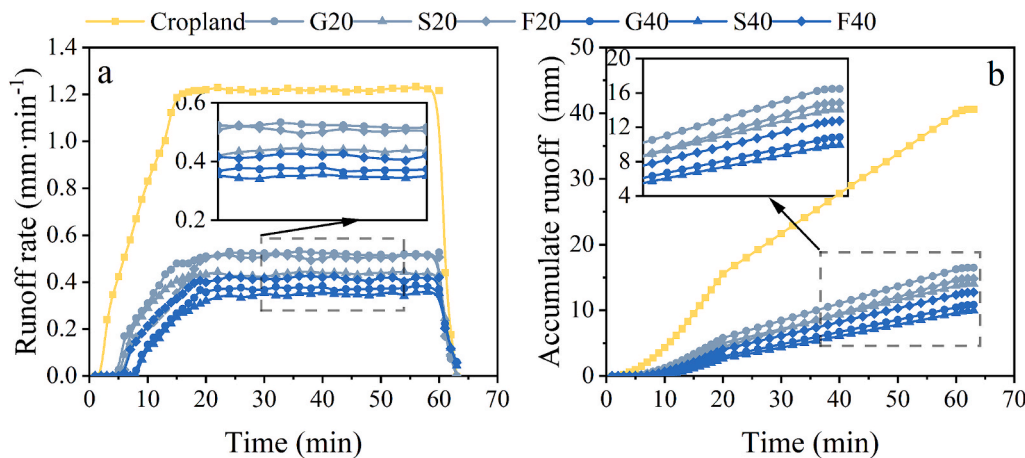


Fig. 3. (a) Variation characteristics of runoff with rainfall time in different experimental plot, (b) cumulative runoff changes with vegetation slope. C represents cropland, G20 represents 20-year grassland, G40 represents 40-year grassland, S20 represents 20-year shrubland, S40 represents 40-year shrubland, F20 represents 20-year forestland, F40 represents 40-year forestland.

Table 2

Characteristics of runoff in different vegetation restoration types.

Vegetation restoration types	IRT (s) Mean (SD)	IRT group	CR (mm) Mean (SD)	CR group	PRT (s) Mean (SD)	PRT group	PCR (mm) Mean (SD)	PCR group
C	<u>142.33</u> (13.47)	c	39.30 (0.78)	a	<u>75.00</u> (5.71)	d	0.62 (0.13)	a
G20	313.67 (20.85)	b	16.23 (0.85)	b	88.33 (3.39)	c	<u>0.24</u> (0.06)	b
S20	364.33 (56.90)	b	13.69 (0.74)	cd	123.67 (11.44)	abc	<u>0.34</u> (0.04)	b
F20	370.67 (33.49)	b	14.57 (0.30)	c	105.33 (15.11)	bcd	0.26 (0.05)	b
G40	487.00 (24.12)	a	10.11 (0.73)	e	146.33 (24.69)	a	0.40 (0.06)	b
S40	522.33 (23.63)	a	<u>9.59</u> (0.28)	e	131.33 (14.26)	ab	0.38 (0.03)	b
F40	382.00 (19.03)	b	12.39 (0.58)	d	96.33 (14.01)	bcd	0.38 (0.07)	b
P values		< 0.001		< 0.001		0.002		0.002

IRT represents initial runoff time, CR represents cumulative runoff, PRT represents post-rainfall runoff time, and PCR represents post-rainfall cumulative runoff. Outside the brackets is the mean, and within the brackets is the standard deviation. Bold numbers indicate the maximum value within the same column, while underlined numbers represent the minimum value within the same column. Different letters in the same column indicate a significant difference at the $P < 0.05$ level.

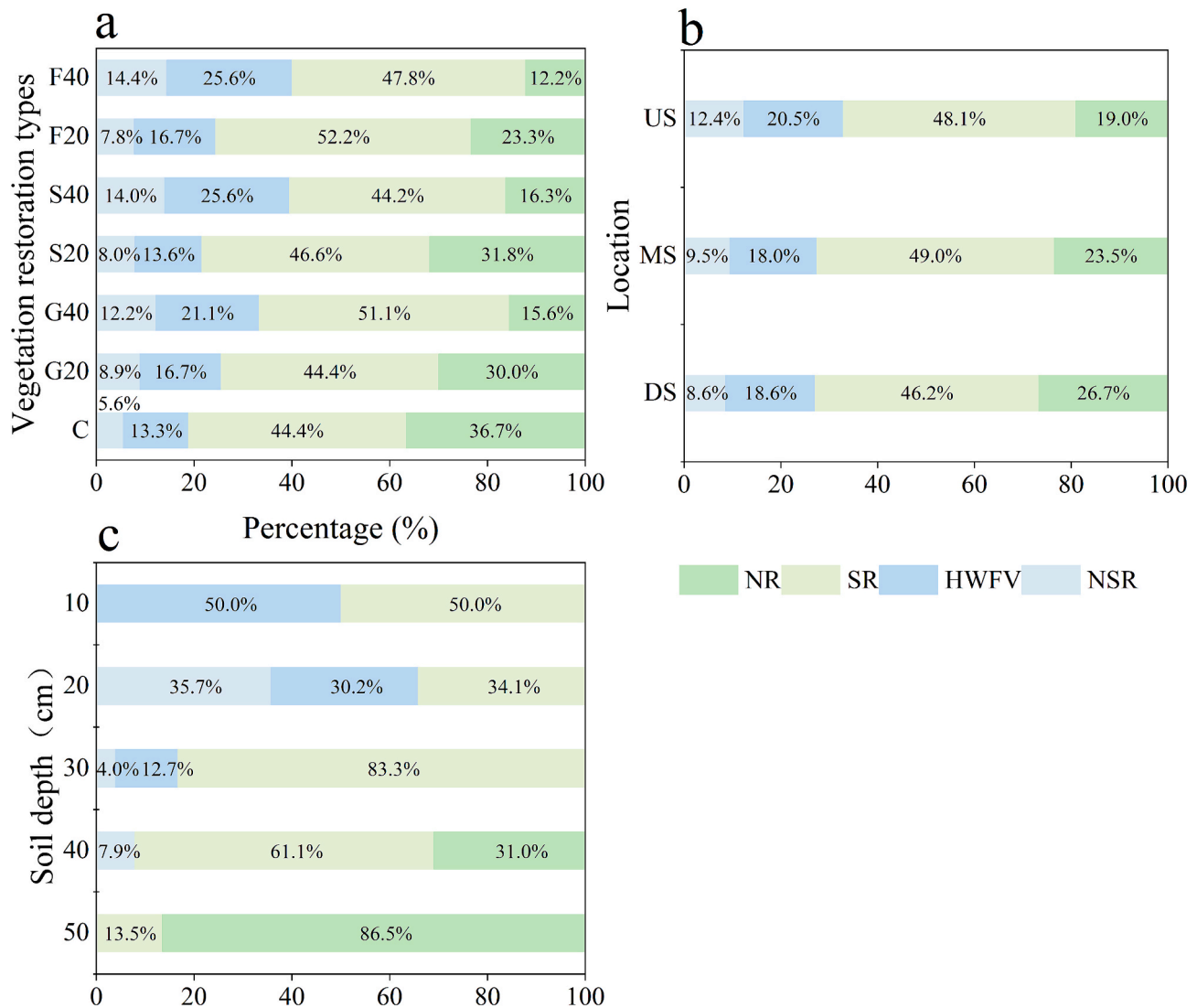


Fig. 4. The occurrence of different infiltration response types in (a) different vegetation restoration types, (b) different slope locations, US represent upslope, MS represent mesoslope, DS represent downslope. (c) different soil depth. NSR: non-sequential response type; HWFV: high wetting-front velocity type; SR: sequential response type; NR: non-response type.

response patterns and vegetation restoration types on soil moisture recession followed the sequence: high wetting-front velocity (HWFV) > non-sequential response (NSR) > sequential response (SR) for patterns, and 40-year grassland (G40) > 40-year forestland (F40) > 40-year shrubland (S40) > 20-year shrubland (S20) > 20-year forestland (F20) > 20-year grassland (G20) > cropland for vegetation types (Fig. 7). With

respect to infiltration events identified by different soil moisture response patterns, the HWFV, NSR, and SR significantly reduced soil moisture by 10.10%, 7.01%, and 6.03%, respectively (Fig. 7a). With respect to vegetation restoration types, soil moisture was significantly lower in revegetated land than in cropland ($P < 0.05$), with soil moisture recession increasing with restoration age (Fig. 7b). Specifically, G40

Table 3

Information of PF infiltration parameters in different vegetation restoration plots.

Land use types	CI-MF (mm)	CI-PF (mm)	CIP-PF (%)	IIR-MF (mm·min ⁻¹)	IIR-PF (mm·min ⁻¹)	IIRP-PF (%)	SIR-MF (mm·min ⁻¹)	SIR-PF (mm·min ⁻¹)	SIRP-PF (%)
C	12.24(0.61)b	18.03(0.19)c	59.59(1.31)b	0.83(0.05)ab	0.91(0.06)b	52.3(2.61)b	0.14(0.01)b	0.19(0.03)c	56.63(2.10)b
G40	16.88(1.02)a	50.58(1.02)b	74.98(1.51)a	0.95(0.03)a	1.05(0.05)b	52.33(1.70)b	0.21(0.02)a	0.71(0.05)b	77.1(1.68)a
S40	17.80(1.74)a	57.37(2.53)a	76.29(2.58)a	0.96(0.11)a	1.03(0.14)b	51.91(5.53)b	0.24(0.02)a	0.82(0.02)a	77.34(1.72)a
F40	14.08(2.36)ab	52.14(3.52)b	78.70(3.60)a	0.67(0.11)b	1.33(0.13)a	66.50(5.31)a	0.20(0.04)ab	0.73(0.08)ab	78.55(4.67)a
P values	0.03	< 0.01	< 0.01	0.02	0.01	0.02	0.02	< 0.01	< 0.01

CI-MF represents the cumulative infiltration of matrix flow; CI-PF represents the cumulative infiltration of preferential flow; CIP-PF denotes the proportion of cumulative infiltration of preferential flow. IIR-MF refers to the initial infiltration rate of matrix flow, while IIR-PF indicates the initial infiltration rate of preferential flow; IIRP-PF represents the proportion of the initial infiltration rate of preferential flow. SIR-M denotes the stable infiltration rate of matrix flow, SIR-PF refers to the stable infiltration rate of preferential flow, and SIRP-PF indicates the proportion of the stable infiltration rate of preferential flow. Outside the brackets is the mean, and within the brackets is the standard deviation. Different letters in the same column indicate a significant difference at the $P < 0.05$ level.

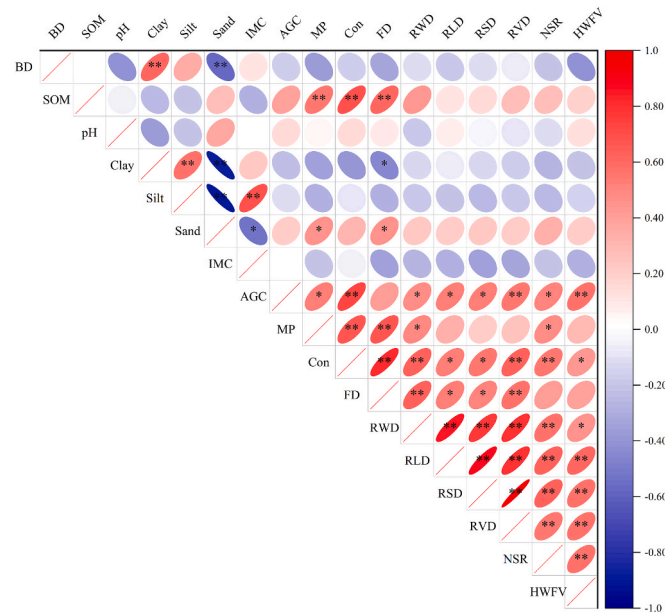


Fig. 5. Correlation analysis of NSR and HWFV with soil physical, chemical properties, and root characteristic parameters (Pearson). * indicates a significant difference at the 0.05 level, ** indicates a significant difference at the 0.01 level. BD: bulk density; SOM: soil organic content; IWC: initial water content; AGC: the content of aggregates larger than 0.25 mm; MP: Soil macroporosity; Con: macropore connectivity; FD: fractal dimension; RLD: root length density; RSD: root surface area density; RVD: root volume density; RWD: root weight density; NSR: non-sequential response; HWFV: high wetting-front velocity.

experienced the greatest recession in soil moisture (8.43%) and cropland the least (5.09%). With respect to revegetated lands other than cropland and F20, soil moisture recession was significantly greater for the HWFV than for the NSR and SR ($P < 0.05$) (Fig. 7c). Among G20, G40, and F40, the NSR resulted in a greater recession in soil moisture than the SR did. Conversely, in S20 and S40, compared with the NSR, the SR resulted in a greater recession in soil moisture. Overall, vegetation restoration significantly enhanced the impact of the HWFV on soil moisture recession.

Among the different infiltration patterns, the recession constant (K) decreased in the order of $NSR > SR > HWFV$ (Fig. 8). These findings indicate that the HWFV significantly accelerated soil moisture depletion and that NSR slowed moisture loss. In terms of vegetation restoration type, cropland presented the highest K values ($K = 0.9770, 0.9154$, and 0.9636), whereas the other revegetated land types presented lower K values, particularly F40, which presented the lowest K values ($0.9111, 0.8916$, and 0.9043 , respectively). This trend suggests that vegetation restoration significantly increased the rate of soil moisture depletion, which intensified over time as vegetation restoration progressed.

Furthermore, the data point distribution in Fig. 8 shows that the NSR has a narrower, more concentrated soil moisture range, indicating a slower recession process. In contrast, the HWFV has a wider, more dispersed soil moisture range, reflecting a faster recession process. We hypothesise that a NSR is more likely to occur during the early stages of rainfall, when soil moisture is relatively low, whereas an HWFV typically occurs during the later stages when the soil is more saturated.

3.5. The relationship between slope runoff and PF

The occurrence of different preferential flow (PF) types differentially affected slope runoff characteristics (Table 4). The results show that PF occurrence was significantly positively correlated with the initial runoff time, post-rainfall runoff time, initial infiltration rate, and stable infiltration rate ($P < 0.05$) and significantly negatively correlated with cumulative runoff ($P < 0.05$). NSR occurrence was significantly positively correlated with the stable infiltration rate ($P < 0.05$) and significantly negatively correlated with cumulative runoff ($P < 0.05$). High wetting-front velocity occurrence followed a similar pattern, with significant positive correlations with initial runoff time, post-rainfall runoff time, initial infiltration rate, and stable infiltration rate ($P < 0.05$) and a significant negative correlation with cumulative runoff ($P < 0.05$). The difference in response time between adjacent soil layers was significantly negatively correlated with the initial runoff time and stable infiltration rate ($P < 0.01$) and significantly negatively correlated with the post-rainfall runoff time and initial runoff time ($P < 0.05$) but being significantly positively correlated with the cumulative runoff ($P < 0.05$). No significant differences were observed between the V_{max} and the runoff characteristics. Overall, vegetation restoration increases PF occurrence, decreases the difference in response time between adjacent soil layers, increases the initial infiltration rate and stable infiltration rate, delays the initial runoff time, reduces cumulative runoff, and prolongs the post-rainfall runoff time.

4. Discussion

4.1. Effects of vegetation restoration on slope runoff and infiltration

Long-term vegetation restoration on China's Loess Plateau has significantly altered soil structure, infiltration capacity, and runoff characteristics across slopes, primarily because of land use changes (Kang et al., 2001; Li et al., 2009; Liao et al., 2023). By using artificial rainfall simulations in the field, this study quantified runoff and infiltration characteristics, and revealed that vegetation restoration generally extended the initial runoff time, reduced cumulative runoff, and increased soil infiltration. Greater vegetation cover amplified these effects (Figs. 5 and 6 and Table 2), which is consistent with the findings of Gu et al. (2019), who noted that this occurred mainly because vegetation restoration increased vegetation cover and litter layer thickness. The canopy, roots, and litter reduce raindrop kinetic energy and promote surface roughness, thereby effectively intercepting slope runoff.

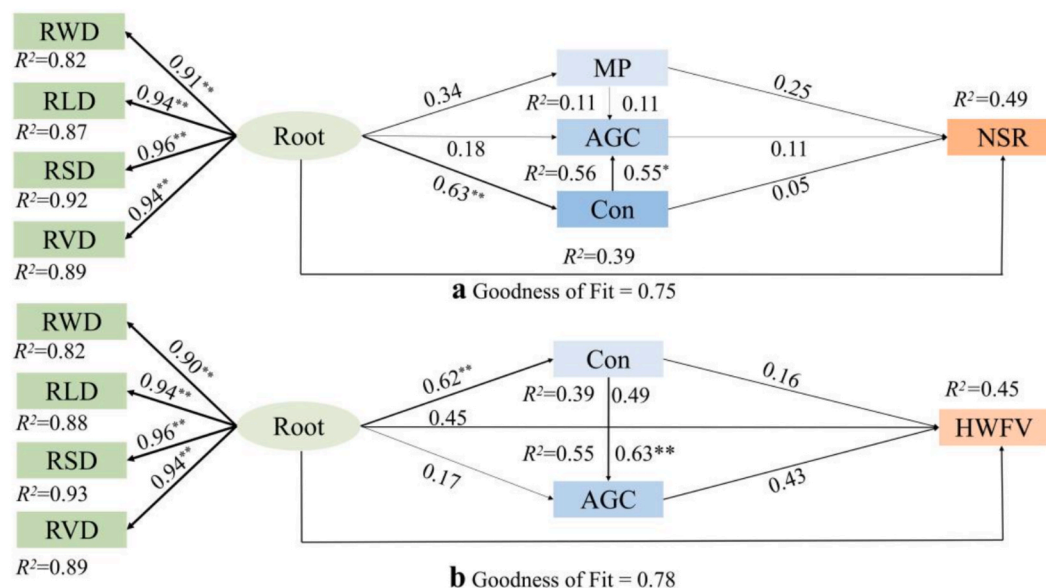


Fig. 6. The partial least squares path modelling (PLS-PM) showed the effects of soil physical properties, chemical properties, and root characteristic parameters on (a) NSR, (b) HWFV. The line width represents the strength of the causal relationship, which is consistent with the standardized path coefficient and appears near the line. R^2 represents the variance of the dependent variable explained by the model. The coefficients of * and ** are significant at the probability levels of 0.05 and 0.01, respectively. The Goodness of Fit statistics are shown below the modeling framework.

This process prolongs the initial runoff time and reduces cumulative runoff. Our findings in this study are consistent with those reported by Zhao et al. (2014a). Herein, a power-law relationship was identified: vegetation cover was significantly positively correlated with initial runoff time, post-rainfall runoff time, initial infiltration rate, and stable infiltration rate and negatively correlated with cumulative runoff and post-rainfall cumulative runoff (Fig. S3). Previous studies have indicated that the post-rainfall cumulative runoff and cumulative runoff decrease with increasing vegetation cover (Vasquez-Mendez et al., 2010), which is consistent with our findings (Fig. S3c, f). This power-law relationship suggests that the effect of vegetation cover on runoff reduction has a threshold beyond which its efficiency decreases. In our study, the effect of vegetation cover on cumulative infiltration also reached a threshold of approximately 50–60%, which aligns with the findings of Jabro (1992) and Gu et al. (2019). However, the coefficient of determination (R^2) for the post-rainfall cumulative runoff and post-rainfall runoff time equation was relatively low (Fig. S3b, c), likely because post-rainfall cumulative runoff and post-rainfall runoff time are influenced by the litter layer, which varies according to the vegetation type (Zhao et al., 2014b). Additionally, the correlation between vegetation cover and the initial infiltration rate was relatively weak ($R^2 = 0.5126$; Fig. S3d), mainly because initial infiltration rate depends not only on vegetation cover but also on soil properties such as the bulk density, porosity, initial moisture content, and soil organic content (Chen et al., 2018) (Fig. S4).

4.2. PF types and influencing factors

In this study, preferential flow (PF) occurrence was identified on the basis of two soil moisture response-based criteria, namely, the non-sequential response (NSR) and high wetting-front velocity (HWVF), with PF occurrence identified by the HWVF occurring 1.72 to 2.40 times more frequently than that identified by the NSR, which is consistent with the work by [Zhang et al. \(2022a\)](#) (Fig. 4). Notably, despite the significant increase in PF occurrence following vegetation restoration, non-preferential flow (i.e., sequential response and non-response types) remained the dominant infiltration pattern, accounting for 60.0–81.1% of total observations across all the plots. This dominance of non-preferential infiltration likely reflects the combined effects of rainfall characteristics, soil hydraulic properties, and limitations associated with

sensor resolution (Nimmo & Perkins, 2018). Our findings indicate that, in addition to plant roots, macropores are the primary factor influencing the occurrence of the NSR (Fig. 6a). This is because soil water can bypass only the soil matrix and trigger a faster response in the lower soil moisture sensors than in the upper ones by flowing through one or more macropores. Therefore, the number and morphology of isolated macropores are key factors affecting the occurrence of the NSR (Zhang et al., 2022a; Qiu et al., 2023). In contrast, for the HWFV, in addition to plant roots, the dominant influencing factors are the content of soil aggregates and the degree of pore connectivity (Fig. 6b). This is because for water to move between soil layers at a rate exceeding the saturated hydraulic conductivity, sufficient connectivity of interlayer pores must be ensured. Soil aggregates provide the most favourable structure for the formation of connected macropores (Bresson & Moran, 2004; Zhao et al., 2017). Previous studies have shown that the content of soil aggregates is significantly positively correlated with the presence of macropores associated with aggregates (Andrews et al., 2002). Our study also suggested that high rainfall intensity is more likely to trigger the HWFV because of the increased positive pressure potential in the surface soil (Gjettermann et al., 1997). The Loess Plateau is characterised by Hortonian overland flow mechanisms (Doody et al., 2010), where high rainfall intensity increases soil erodibility. The transport of fine particles by runoff can clog surface and near-surface soil pores, potentially inhibiting the NSR. However, the HWFV may still occur in sediment-filled macropores, which serve as interconnected conductive porous media (Nimmo, 2021).

In this study, PF occurrence was relatively insensitive to slope location (Fig. 4 and Table S1). This is likely because the slope gradient alone cannot fully capture topographical elements directly related to PF behaviour, such as microtopography and surface roughness (Graham & Lin, 2011). Interestingly, a weak downslope decrease in PF frequency was observed, which is consistent with the findings of previous studies (Liu and Lin, 2015; Wang et al., 2018; Chen et al; 2021). This pattern may be partly explained by the downslope transport of fine particles during prolonged rainfall events, which can lead to partial blockage of PF pathways in surface soils at lower slope locations and thereby constrain PF development and intensity (Budhathoki et al., 2022). In contrast, Mei et al. (2018) reported that on semiarid loess slopes, the accumulation of rock fragments on lower slopes resulted in greater PF

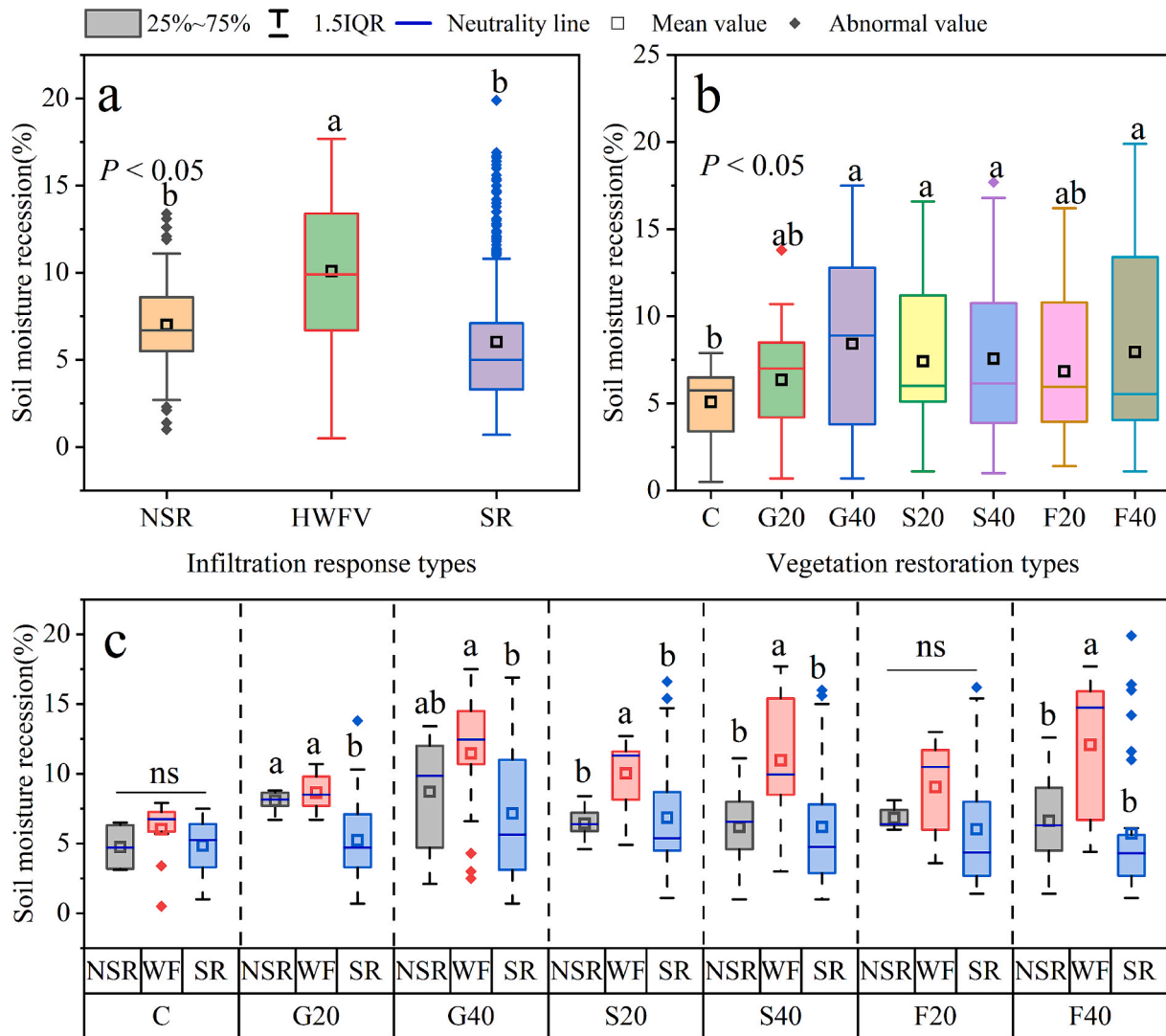


Fig. 7. Distribution of soil moisture recession in (a) different infiltration response types, (b) different vegetation restoration types, (c) interaction between vegetation restoration types and infiltration response types. Different lowercase letters indicate that there are significant differences under different infiltration response types or vegetation restoration types ($P < 0.05$), ns showed no significant difference. C represents cropland, G20 represents 20-year grassland, G40 represents 40-year grassland, S20 represents 20-year shrubland, S40 represents 40-year shrubland, F20 represents 20-year forestland, F40 represents 40-year forestland.

occurrence and greater infiltration variability. These findings suggest that slope location alone does not fully explain PF patterns and that other factors, such as soil type or structure, may also influence PF pathways.

Land use is a crucial factor that controls water flow and significantly influences soil moisture response processes (Bachmair et al., 2009). Our results revealed that vegetation restoration significantly increased PF occurrence (Fig. 4a). The conversion from cropland to revegetated land (i.e., shrubland, forestland or grassland) drives rainfall redistribution through the canopy, stem and litter layers, thereby reducing the splashing effect of raindrops and promoting the development of PF by guiding water to the soil near the stem base (Johnson & Lehmann, 2006). Moreover, vegetation cover can increase soil organic content, which enhances soil hydrophobicity and soil faunal activity (e.g., earthworms and rodents), creating a favourable environment for PF (Gjettermann et al., 1997; Nespoulous et al., 2019; Gan et al., 2024; Wen et al., 2025). Our study also revealed that PF occurrence in forestland was greater than that in grassland and shrubland (Fig. 4a). This is likely because compared with those in grasslands and shrublands, tree roots in forestland are thicker, longer, and more interconnected. As the roots decompose, they form more complete pathways for PF, which is

consistent with the studies of Tian et al. (2023) and Li et al. (2023). However, these results differ from those of Luo et al. (2019) and Cui et al. (2019), who reported that coarse roots contribute less to PF transport than fine roots do, which constitute a much larger proportion of the total root system in the soil profile. The partial least squares path modelling results revealed that compared with the root length density and root weight density, the root surface area density and root volume density had greater influences on PF development (Fig. 5). Coarse roots can increase the radius of root channels and increase water infiltration (Ghestem et al., 2011; Bodner et al., 2014). The thicker the roots are, the greater the root surface area and volume. Therefore, compared with the root length density and root weight density, the root surface area density and root volume density are more conducive to PF. Additionally, PF occurrence increased with revegetation age (Fig. 4a), a finding consistent with that of Cai et al. (2024). Cheng et al. (2014) also reported significant vertical differences in PF across different forest ages, with the depth of PF increasing with forest age. This may be attributed to the continuous development and extension of root systems during vegetation restoration, which promotes PF (Kan et al., 2020).

The correlation results indicated that the content of aggregates larger than 0.25 mm, macroporosity, connectivity, and root system

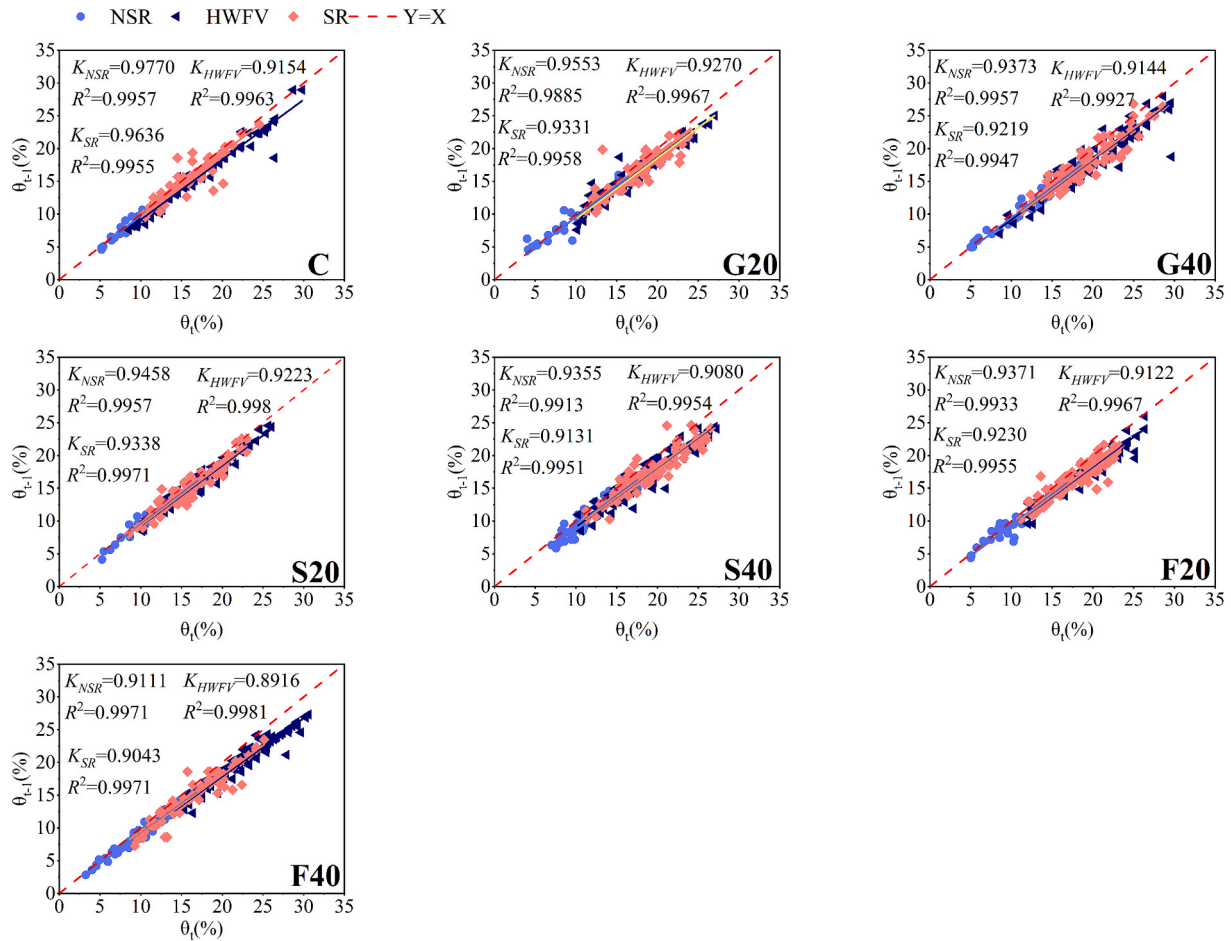


Fig. 8. Recession constant (K) of NSR, HWFV, and SR for different vegetation restoration types. C represents cropland, G20 represents 20-year grassland, G40 represents 40-year grassland, S20 represents 20-year shrubland, S40 represents 40-year shrubland, F20 represents 20-year forestland, F40 represents 40-year forestland.

Table 4
Pearson correlation analysis of runoff characteristic value and PF.

Runoff characteristic value	TF	NSR	HWFV	DRT	V_{max}
IRT	0.49*	0.37	0.48*	-0.61**	-0.06
CR	-0.53*	-0.50*	-0.45*	0.58**	-0.26
PRT	0.46*	0.3	0.49*	-0.46*	-0.16
PCR	-0.07	-0.1	-0.04	-0.21	-0.40
IIR	0.46*	0.28	0.50*	-0.48*	0.12
SIR	0.55**	0.54*	0.47*	-0.56**	0.30

TF represents the sum of NSR and HWFV frequencies. *: $P < 0.05$; **: $P < 0.01$.

characteristics (root length, surface area, volume, and weight density) significantly affected PF occurrence (Fig. 5). The results from the PLS-PM revealed that root system characteristics were the most direct and critical factors affecting both the NSR and HWFV (Fig. 6). As an essential component of vegetation and soil, roots act as a vital medium for PF (Bourrioug et al., 2015). Under the combined influences of plant root growth and development, soil faunal activity, wetting-drying cycles, and freeze-thaw cycles, the soil develops a variety of morphologically distinct and locally interconnected pores or fractures, which provide favourable pathways for the initiation and development of PF (Cai et al., 2024). Our findings revealed significant positive correlations between increased PF occurrence and increased root length, surface area, volume, and weight density ($P < 0.01$). This effect arises from the complex root networks formed by vegetation, which improve soil structure and hydraulic properties (Shi et al., 2021), thus enhancing soil infiltration. Plant roots not only physically bind soil particles but also release

exudates that increase the soil organic content and the content of aggregates larger than 0.25 mm. The size, distribution, and stability of the content of aggregates larger than 0.25 mm influence hydrological properties, such as soil saturated hydraulic conductivity (Cui et al., 2019). Moreover, root decay promotes the formation of macropores, further facilitating NSR. Studies have shown that soils containing roots exhibit distinct physical and chemical properties as well as higher active water infiltration behaviour than soils without roots do (Jiang et al., 2017). Liao et al. (2023) investigated the infiltration capacity of different land use types on the Loess Plateau. They reported that root systems were a major factor influencing soil infiltration rates, with infiltration rates positively correlated with root biomass. Furthermore, stable macropores formed by both living and decaying roots promote the development of PF in deeper soil layers (Cui et al., 2019; Li et al., 2023). These results underscore the crucial role of root systems in PF formation and development.

4.3. Effects of PF on infiltration and slope runoff

Soil infiltration is a fundamental hydrological process in terrestrial ecosystems that partitions rainfall into surface runoff and infiltrating water (Vlcek et al., 2017; Lozano-Baez et al., 2019; Kidron, 2021), establishing connections among groundwater, surface water, and soil moisture. Preferential flow (PF) impacts infiltration capacity, increasing the complexity and unpredictability of water redistribution in the soil (Allaire et al., 2009). Although PF pathways constitute only a small fraction of the total soil volume, they can contribute approximately

11–94% of the total infiltration (Alaoui, 2015). Our study revealed that cumulative PF infiltration accounted for 59.59–78.70% of the total infiltration (Table 3). Additionally, the PF contribution in the F40, S40, and G40 plots was significantly greater than that in the cropland plots ($P < 0.01$) because of the deeper and more complex root systems in forestland and shrubland than in cropland or grassland. Our results also revealed that the stable infiltration rate of PF was 1.36–3.42 times greater than that of matrix flow (MF) (Table 3). Similarly, Zhang et al. (2019a) reported that stable infiltration rates of PF values were 2–5 times higher than those of MF values, supporting our findings. Conversely, the findings of Sanders et al. (2012) from neighbouring areas within the Loess Plateau revealed that surface-connected macropores accounted for 99% of the total infiltration. Additionally, Zhang et al. (2022b) used in situ double-ring infiltrometer tests and laboratory infiltration experiments and reported that the stable infiltration rates of PF were 34–118 times higher than those of MF, with PF contributing substantially more to total infiltration than in our study. This discrepancy may stem from the use of remoulded soil samples to measure matrix infiltration rates in the laboratory, which differ from those of natural soil samples, and from overlooking MF–PF interactions under natural conditions. Using field rainfall experiments combined with time-lapse electrical resistivity tomography, Litt et al. (2020) quantified the impact of different land uses (land cover) on soil infiltration capacity. They concluded that PF significantly enhanced infiltration, particularly in older forest stands. Our study also revealed that with increasing revegetation age, the soil infiltration capacity significantly increased, and the contribution of PF to total infiltration became more pronounced (Table 3). Similar findings have been reported by Gu et al. (2019), Zhao et al. (2023), and Cai et al. (2024).

Slope runoff, a fundamental component of catchment runoff, plays a crucial role in watershed hydrological processes. On the Loess Plateau, restored vegetation and reduced human disturbance have modified root characteristics and surface litter, altering the soil structure and changing PF patterns and pathways, which significantly impact slope runoff (Li et al., 2009). Our study revealed that the occurrence of NSR was positively and negatively linearly correlated with both cumulative runoff and the stable infiltration rate ($P < 0.05$), whereas a high wetting-front velocity (HWFV) was positively correlated with cumulative runoff, initial runoff time, post-rainfall runoff time, initial infiltration rate, and the stable infiltration rate ($P < 0.05$) (Table 4 and Fig. S5). This may be due to the more sporadic and heterogeneous nature of non-sequential response (NSR), which is closely related to macropore distribution. Conversely, compared with the NSR, a HWFV often occurs in well-structured soil layers, which explains its greater occurrence (Fig. 4) and its significantly greater impact on soil moisture recession (Figs. 7 and 8). These findings are supported by those of previous studies (Zhu et al., 2008; Zhang et al., 2022a).

Our integrated analyses identify three synergistic pathways through which PF reduces overland flow. First, macropore-mediated rapid infiltration enables water to bypass the soil matrix at rates 1.36–3.42 times higher than those of MF (Table 3), particularly through the NSR (Beven and Germann, 2013). Second, root-enhanced infiltration networks create continuous three-dimensional pathways that sustain high-velocity water movement, predominantly driving HWFVs (Bodner et al., 2014). Third, aggregate-stabilised infiltration maintains persistent pore structures, ensuring stable infiltration rates of 0.52–0.82 mm·min⁻¹ in revegetated land (Bresson and Moran, 2004). Compared with cropland, vegetation restoration amplifies all three mechanisms simultaneously through root development, enhanced pore connectivity, and organic matter accumulation, resulting in a 58.70–75.60% reduction in cumulative runoff (Table 2). These pathways collectively shift rainfall-runoff relationships from infiltration-excess-dominated overland flow to infiltration-dominated water partitioning in restored ecosystems (Dunne and Black, 1970).

4.4. Implications for vegetation management and research limitations

As vegetation restoration progresses on the Loess Plateau, it influences hydrological processes in two primary ways: (1) interception by the canopy, litter layer, and roots, and (2) improvement of the soil structure (Kalhor et al., 2018). Changes in soil properties and root characteristics following the conversion of cropland to forestland, shrubland, and grassland are shown in Fig. 9. Vegetation restoration promotes PF development and enhances infiltration capacity, further reducing surface runoff and increasing soil infiltration and water retention by plants (Yu et al., 2015). Given that rainfall during the rainy season is the only source of soil moisture on the Loess Plateau (Jia et al., 2017), increasing the storage and efficient use of this limited resource is critical for ecological restoration.

Currently, the artificial forests on the Loess Plateau are relatively homogeneous, and dominated by *Robinia pseudoacacia*, which increases the risk of large-scale pest outbreaks and results in limited ecological functions (Zhang et al., 2023b). Reports of forest dieback caused by drought and pest infestations on the Loess Plateau are widespread and require urgent attention. Our results revealed that forested areas significantly increased PF occurrence, subsequently improving infiltration capacity. However, PF also accelerated soil moisture depletion (Figs. 7 and 8), exacerbating evaporation. The well-developed macropore networks that facilitate rapid infiltration during rainfall also create efficient pathways for vapour diffusion during dry periods, enhancing evaporation from deeper soil layers (Nachshon et al., 2011). Second, vegetation restoration significantly increases plant water consumption through transpiration, with mature forests (*Robinia pseudoacacia*) consuming 400–600 mm annually on the Loess Plateau—approaching or exceeding the mean annual precipitation in dry years (Wang et al., 2012; Jia et al., 2017). Third, the deeper infiltration enabled by PF places water below the root zone of shallow-rooted crops but within the reach of deep-rooted trees and shrubs, making this water accessible for plant uptake but unavailable for shallow soil moisture storage (Chen et al., 2007). Some studies suggest that semi-shrubs and naturally restored grasslands, which require less water, may be more suitable for restoration in plateau regions (Wang et al., 2012). Qiu et al. (2024) recommended planting trees and shrubs on the upper sections of steep slopes and grass on the lower sections to reduce tree and shrub density on the lower slopes and maintain an optimal vegetation balance. In summary, successful vegetation restoration on the Loess Plateau requires careful consideration of vegetation type, slope position, soil depth, and soil properties, all of which influence PF dynamics, to achieve sustainable ecological restoration.

Previous studies have used soil moisture sensors and macropore-blocking methods to quantify PF occurrence and contribution, validating these approaches (Koestel & Jorda, 2014; Zhang et al., 2019a; Zhang et al., 2022b; Guan et al., 2024). Despite the valuable and innovative findings of our study, some limitations exist within the plot experiment. First, although we minimised disturbances by placing soil moisture probes outside the runoff plots, their insertion inevitably disturbed the soil within the plots, potentially creating artificial macropores or cracks that may have influenced PF occurrence. To mitigate this effect, we inserted the probes in a single motion and compacted the surrounding soil to minimise experimental interference. Additionally, when assessing PF contribution, we assumed a uniform soil structure across the three runoff plots, not accounting for soil heterogeneity. Although this assumption may be valid for uniform soils, further verification is needed for heterogeneous soils. Furthermore, the sand–soil mixture used to block surface PF pathways may not have fully sealed these pathways; therefore, future studies should consider alternative sealing materials. Finally, the small size of our runoff plots may have restricted our ability to capture high-frequency PF events. Future research should expand the plot size and increase the spatial density of soil sensor monitoring.

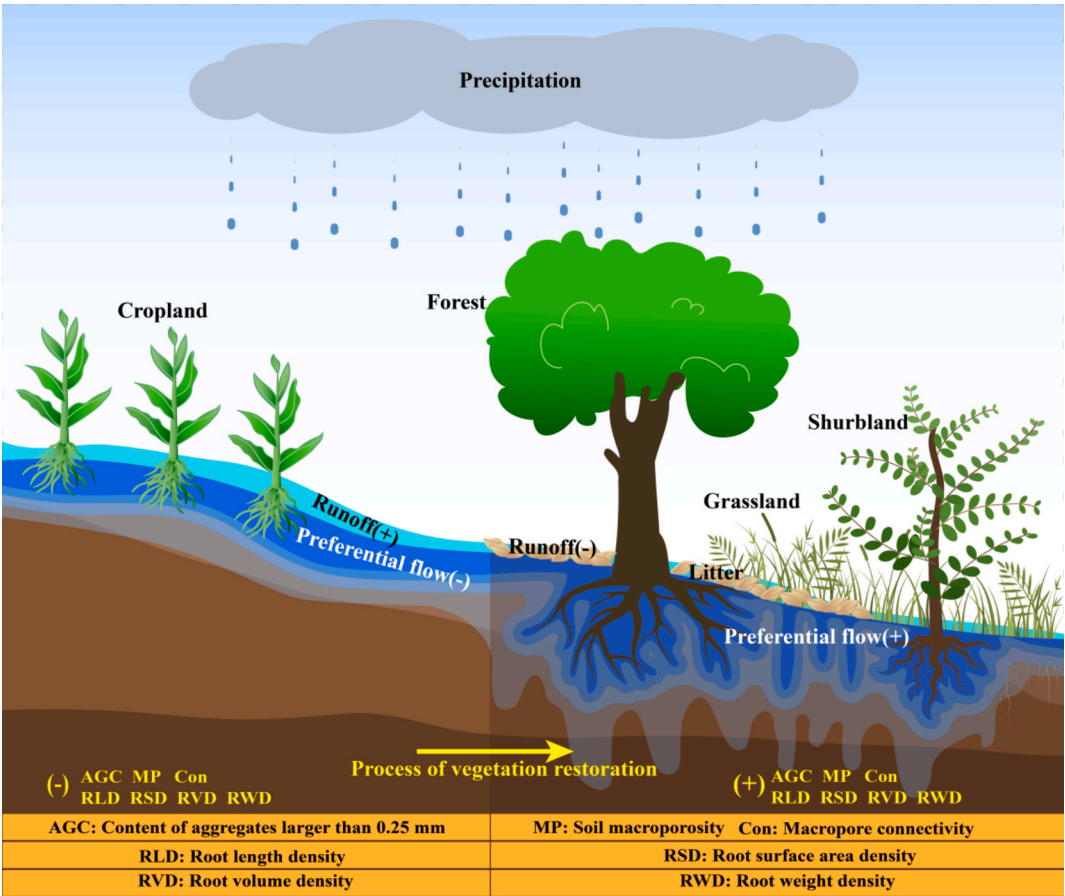


Fig. 9. Conceptual diagram of changes in soil properties, root characteristics, and hydrological processes after natural vegetation restoration (conversion of farmland to forestland, shrubland, and grassland). vegetation restoration increases the AGC, MP, Con, RLD, RSD, RVD, and RWD. These factors enhance the PF and infiltration capacity, while further reducing surface runoff.

5. Conclusion

In this study, the occurrence and proportion of preferential flow (PF) under different vegetation restoration types on the Loess Plateau were quantified using field artificial rainfall experiments combined with soil moisture sensor monitoring and a surface covering method. The results indicated that vegetation restoration significantly reduced runoff rates and increased soil infiltration, with greater revegetation age leading to lower runoff and higher infiltration rates. Compared with cropland, revegetated land resulted in significant increases in the cumulative infiltration, initial infiltration rate, and stable infiltration rate of both PF and matrix flow (MF) ($P < 0.05$). PF occurrence was identified on the basis of two soil moisture response-based criteria, namely, non-sequential response (NSR) and high wetting-front velocity (HWVF), with PF occurrence identified by HWVF occurring 1.72–2.40 times more frequently than that identified by NSR. The NSR was positively correlated with cumulative infiltration and negatively correlated with the stable infiltration rate ($P < 0.05$), whereas the HWVF was significantly negatively correlated with cumulative infiltration and post-rainfall cumulative runoff ($P < 0.05$) and significantly positively correlated with initial runoff time, initial infiltration rate, and stable infiltration rate ($P < 0.05$). PF occurrence was significantly greater in forestland, shrubland, and grassland than in cropland, with greater revegetation age associated with increased PF occurrence. PF occurrence was highest on the upper slope (32.86%), followed by the mesoslope (27.50%) and downslope (27.14%), although these differences were not significant ($P > 0.05$). PF occurrence significantly decreased with increasing soil depth ($P < 0.01$). A significant positive correlation was observed between PF occurrence and the proportion of cumulative infiltration of PF

($R^2 = 0.69$, $P < 0.01$), with PF contributing between 59.59% and 78.70% of the total infiltration. Root characteristics had the strongest and most direct influence on the NSR, followed by soil macroporosity and the content of aggregates larger than 0.25 mm. With respect to the HWVF, root characteristics were also the most influential factor, followed by the content of aggregates larger than 0.25 mm and macropore connectivity. Vegetation restoration influences slope infiltration and runoff generation by increasing PF occurrence and its contribution to infiltration. This research enhances the understanding of the effects of vegetation restoration on PF, infiltration, and runoff on the Loess Plateau, offering insights into changes in surface runoff mechanisms under vegetation restoration.

CRediT authorship contribution statement

Yongfu Wen: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Chaojun Gu:** Writing – review & editing, Methodology, Conceptualization. **Peng Gao:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **Jiahui Zhou:** Writing – review & editing, Methodology, Conceptualization. **Xiaoxue Guo:** Writing – review & editing, Methodology, Conceptualization. **Xingmin Mu:** Writing – original draft, Supervision, Project administration. **Xucheng Ai:** Methodology, Investigation, Conceptualization. **Ruidun Zhang:** . **Siyu Ren:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135241>.

Data availability

Data will be made available on request.

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